PROJECT-BASED LEARNING

PHASE-II EVALUATION

UNIGO

Team ID: T159

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T159
Semester: 3rd
Section: ML1,ML2,ML3,DS2
Domain: *AI-ML,AI-DS*

Team Members

1. *Sapna Yadav – 2510011587*
2. *Srishti Nahar – 2510014554*
3. *Vanshika – 2510012685*
4. *Anushka Pun – 2512510037*

Mentor

Ms. Megha Garg **Interactions completed:4**

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

Phase-I Recap & Feedback Incorporated

Phase-I Status

UNIGO was proposed as a centralized platform to provide campus navigation, bus information, Lost-Found, and Reviews-Feedback through a single platform.

Feedback Received

- *Completing all four features simultaneously was too broad.*
- *Timetable data was essential for the verification of bus information.*
- *A search was required to handle partial names and typos with a modular design for upcoming functionalities.*

Changes Incorporated

- *Carried out the Bus Finder feature entirely and with an organized plan,82 destinations have been set across four departure points and three shift timings.*
- *Included auto-complete and "Did you mean" suggestions.*
- *Designed a homepage featuring four blocks,with real-time Bus Finder functionality and the other features labelled as "Coming Soon".*

Problem Statement & Finalized Objectives

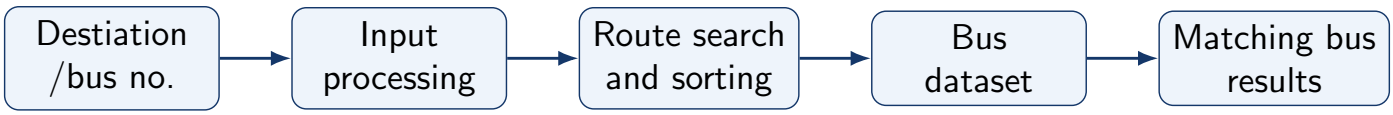
Problem Statement

Students often face difficulty in finding the correct bus, route, intermediate stops, and departure timings. A simple search facility is needed to access this information easily.

Finalized Objectives

1. *Provide the departure location along with the schedule for the three shifts at 82 destinations.*
2. *Ensure that search results are returned in less than a second.*
3. *Handle partial names, different cases, extra spaces, and typo.*
4. *Permit modifications to the schedule without altering the code.*
5. *Ensure that the platform remains modular for the following three features.*

Updated System Architecture / Workflow



Major Components

- *Web Frontend (HTML, CSS, JavaScript).*
- *C++ HTTP Server and Request Handlers.*
- *Bus Search Engine.*
- *Data Store.*

Changes Since Phase-I

- *The separate database server was substituted with an in-memory indexed data store.*
- *Added "Did you mean" suggestions for spelling mistakes.*
- *Allowed automatic data reloading without server restarts.*

Module Design & Functional Requirements

Major Modules

1. *User Interface and Session.*
2. *The Bus Search Engine.*
3. *The Web Server / API.*
4. *Data Management.*

Functional Requirements

- *Remember the student's name until you log out.*
- *Show the pickup spot, three operating schedules, and the next vehicle needed.*
- *Auto-complete and suggestions for alternative terms.*
- *Reject empty or invalid inputs with a clear message.*

Module Dependency

Requests are sent from the User Interface to the Web Server, which in turn forwards them to the Bus Search Engine. The search engine uses the Data Management module to access bus data, which pulls and updates records from the CSV dataset.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++14, HTML, CSS and JavaScript*
- Data storage: *CSV dataset with in-memory indexing*
- Framework: *None; custom HTTP server using Winsock sockets*
- Development tools: *VS Code, MinGW g++ and Python (openpyxl)*
- Version Control: *GitHub*

Development Methodology

1. Detected the bus schedule issue and gathered route information.
2. Established components, interfaces and data frameworks.
3. developed the frontend, search engine, and web server individually.
4. Linking the frontend to the C++ backend was achieved via a JSON API.
5. A total of 52 automated tests and manual checks were conducted.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures	*Hash Table, Trie, Balanced BST (std::map), Heap / Priority Queue, Hash Set, Stack, Dynamic Programming (Edit Distance), Recursion (DFS)*	*Bus Finder-Search engine*
OOPs with C++	*Classes & Objects, Encapsulation, Abstraction, Inheritance, Runtime Polymorphism, Composition, Smart Pointers (RAII), Exception Handling, STL*	*Bus Finder-Search engine and Web Server*

Development & Implementation Progress

Module	Status	Work Completed
Homepage and Name entry	Completed	*Page for initiating action with a Get Started button and field for entering a name, featuring validation.*
Home and Navigation	Completed	*A personalized greeting, featuring four sections: Home, Features, Logout, and About.*
Bus Finder	Completed	*Search for destinations, boarding location, multiple shift options, upcoming bus, auto complete, and misspelling suggestions.*
Additional Features	Pending	*Planned for next phase.*

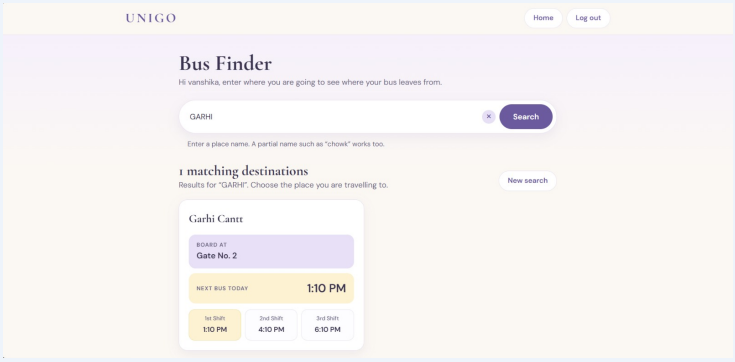
Overall Project Completion: *45%*

Working Prototype / Demonstration

Implemented Features

- *Homepage with your name, with a greeting,then log out.*
- *Homepage with four main sections and additional information in About section.*
- *Bus Finder-pick-up spot,3 operational schedules, and the next vehicle.*
- *Auto-completion along with suggestions for correcting spelling errors.*
- *Clear instructions for empty, invalid, or unrecognized destinations.*

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Search "ISBT"	*Parking near hill(old)*	*Parking near hill(old)*	*Pass*
Search "chowk"	*All matching passes*	*17 results*	*Pass*
Typo "isbtt"	*Suggest ISBT*	*"Did you mean:ISBT*	*Pass*
Mixed case and extra spaces	*Correct place foud*	*Found*	*Pass*
Empty/Invalid input	*Error message*	*Error message shown*	*Pass*
Edit dataset while running	*Updated data shown*	*Updated without restart*	*Pass*

Results / Validation

All 52 automated testing checks executed successfully. According to 100 tests, the average response time for searches was around 2 milliseconds. No errors occurred during the loading of all 82 dataset records, indicating a successful bus data load.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: `https://github.com/Anushkapun150507/PBL_DSCPP_SEM3`
- README.md: *Updated*
- Contributors: *All Members Added*
- Repository Status: *Updated*

Development Activity

- *Commits recorded in the repository.*
- *Contributions from all four team members.*
- *Latest implementation uploaded to GitHub.*
- *Individual commits recorded under contributor accounts.*
- *Project code and README maintained in the repository.*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Sapna Yadav	*Frontend and Backend Search Developer*	*Website interface, bus-search logic, and responsive design, with emphasis on destination matching and search-result handling.*	*22*
Vanshika	*Frontend logic and Data Developer*	*Navigation pages, storage of names, logout functionality, autocomplete, and an Excel to CSV converter*	*22*
Srishti Nahar	*Backend Developer(Web Layer)*	*A C++ HTTP server featuring request and response classes, a handler class structure, and socket operations*	*22*
Anushka Pun	*Backend Developer*	*Data structures such as tries and hash maps, typo correction with edit distance, next-bus calculations, and automated testing.*	*22*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *GCC 6.3's limitations,the compiler did not support C++17 features.*
- *Seamless linking of the website.*
- *Issues with parsing, due to Excel time formats and commas in place names.*

Solutions / Learning

- *C++14, a custom HTTP server with a JSON API.*
- *exact searches-hashing,prefix searches-trie,top results-heap,typo corrections-dynamic programming.*

Phase-III Roadmap

1. *Create the Campus Map utilizing graphs and shortest-path algorithms.*
2. *Build Lost & Found with a database storage system.*
3. *Create Reviews and Tips.*
4. *End-to-end system integration along with multi-threaded server development.*
5. *Conduct user testing, prepare documentation, and proceed with deployment.*

Expected Outcomes & References

Expected Final Outcomes

- *A fully functional platform featuring all four functions.*
- *Quick searching with a response time measured in milliseconds.*
- *Practical application for students, particularly beginners.*
- *Simple timetable modifications for the transport office.*
- *A flexible, component-based layout for upcoming campus offerings.*

References

1. *Research Paper / Journal: E. Fredkin, "Trie Memory," CACM, 1960.*
2. *Book: Cormen et al., Introduction to Algorithms, 3rd ed., 2009.*
3. *Official Documentation: Microsoft Learn (Winsock), cppreference (C++).*
4. GitHub Link: `https://github.com/Anushkapun150507/PBL_DSCPP_SEM3`

Thank You
Questions & Suggestions